

AI-Driven Expense Reimbursement Automation

Phased Delivery Plan

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Approach Overview

The solution is built as a layered pipeline, where each layer is a phase of delivery: Capture → Extract (Claude) → Validate (rules + Claude reasoning) → Detect anomalies → Automate approvals and reimbursement → Integrate with enterprise systems → Surface a self-service employee experience → Pilot and scale. Each phase produces a working, demonstrable increment rather than a big-bang release.

Claude is used in four specific, high-leverage roles rather than as a general-purpose bolt-on: (1) multimodal extraction of receipts into structured data, (2) judgment-based policy reasoning for cases static rules can't resolve, (3) document forensics and contextual anomaly reasoning, and (4) a grounded conversational assistant for employees and approvers. Deterministic rules and ML models handle the high-volume, low-ambiguity work; Claude handles the parts that require reading, judgment, and explanation.

Benchmark: Navan's platform (Expense Agent, Audit Agent, Joule agents) reflects where this category is heading — point-of-transaction policy enforcement, 130+ data points captured automatically per transaction, and AI-generated-receipt detection. This plan targets equivalent capability, built around Claude as the reasoning layer.

Roadmap at a Glance

Phase	Focus	Timeline	Primary Claude Role
0	Discovery, Requirements & Data Landscape	Weeks 1–2	—
1	Receipt Capture & Intelligent Data Extraction	Weeks 3–6	Multimodal receipt/line-item extraction
2	Policy Validation Engine	Weeks 7–10	Judgment-based policy reasoning & explanations
3	Anomaly & Fraud Detection	Weeks 11–14	Document forensics & contextual risk reasoning
4	Workflow Automation & Approvals	Weeks 15–18	Exception routing & approval summaries
5	Enterprise System Integration	Weeks 19–22	—
6	Employee Experience & Analytics	Weeks 23–25	Grounded employee assistant
7	Pilot, Hardening & Rollout	Weeks 26–30	—
8	Continuous Improvement (Post-Launch)	Ongoing, quarterly cadence	Ongoing model/rule feedback review

Total build timeline: ~30 weeks (7 months) to full rollout, followed by an ongoing continuous-improvement cadence.

Phase-by-Phase Plan

Phase 0 — Discovery, Requirements & Data Landscape

Objective	Establish scope, success criteria, and technical foundations before development starts, so every later phase builds on an agreed policy model and system inventory rather than assumptions.
Key Deliverables	- Stakeholder interviews (Finance, Travel, HR, IT/Security) documenting current pain points and policy rules - Inventory of existing systems: ERP/GL (SAP, Oracle, NetSuite, Workday, Dynamics), HRIS, card/payment feeds, and any legacy expense tool - Expense policy rulebook translated into a structured, machine-readable format (per-diem limits, category caps, approval thresholds, prohibited categories) - Data privacy/security requirements (SOC 2, GDPR, PCI where card data is involved) and access model - Baseline metrics: current reimbursement cycle time, error/rejection rate, audit sampling %, employee satisfaction score
Technical Approach	Structured requirements workshops; policy-as-data schema design (JSON/YAML rules engine spec); architecture review of the integration surface (real-time APIs vs. batch/flat-file feeds).
Timeline	Weeks 1–2
Success Metrics	- Signed-off requirements document - Policy rulebook v1 complete and reviewed by Finance - Full integration inventory with owners identified
Key Risks	- Undocumented policy exceptions surface late - Systems of record are fragmented across business units

Phase 1 — Receipt Capture & Intelligent Data Extraction

Objective	Build the ingestion layer that turns receipts — photo, email, PDF, or card feed — into structured, validated transaction data. This is the foundation every later phase depends on, matching the standard set by tools like Navan's Expense Agent, which captures 130+ data points per transaction automatically.
Key Deliverables	- Multi-channel capture: mobile photo upload, email forwarding, PDF/CSV import, direct card-feed ingestion - Claude-powered multimodal extraction of vendor, date, amount, currency, tax, and individual line items — not just totals, so downstream policy checks can validate at line-item level - Duplicate / near-duplicate detection at ingestion (hash plus fuzzy match on vendor, date, amount) - Confidence scoring, with low-confidence extractions routed to a human review queue - A structured transaction data store feeding every downstream phase
Technical Approach	Claude vision + structured output (JSON schema / tool use) for receipt parsing, with a validation

	layer that cross-checks extracted totals against line items to catch extraction errors automatically.
Timeline	Weeks 3–6
Success Metrics	● >95% field-level extraction accuracy on the pilot receipt set ● <5% of receipts routed to manual review ● Average processing time per receipt under 10 seconds
Key Risks	● Handwritten or foreign-language receipts reduce accuracy ● Poor-quality mobile photos ● Multi-currency conversion accuracy at time of expense

Phase 2 — Policy Validation Engine

Objective	Enforce expense policy automatically and in real time, flagging or blocking violations before reimbursement rather than after — mirroring the point-of-transaction enforcement model used by leading tools.
Key Deliverables	● Deterministic rules engine consuming the Phase 0 policy rulebook (category caps, per-diem, receipt-required thresholds, role-based limits, prohibited merchants) ● Claude-based policy reasoning layer for nuanced cases pure rules can't cover (e.g., "is this client dinner reasonable for the deal size and location?") ● Auto-approval path for fully compliant, low-risk expenses; auto-flag path with explainable reason codes for violations ● Plain-language policy violation notifications to submitters, not just a "denied" stamp ● Policy versioning so rule changes don't retroactively break existing audit trails
Technical Approach	A fast deterministic rules engine handles the majority of cases; Claude handles the judgment-based edge cases and generates human-readable explanations for every flag, so both employees and auditors understand the "why."
Timeline	Weeks 7–10
Success Metrics	● 90%+ of expenses auto-adjudicated without human touch ● Policy violation false-positive rate under 5% ● 100% of flagged items carry an explainable reason
Key Risks	● Over-flagging creates employee friction and erodes trust ● Some policy rules resist clean structured-logic representation

Phase 3 — Anomaly & Fraud Detection

Objective	Catch what static rules miss: duplicate submissions, inflated claims, AI-generated or manipulated receipts, and unusual spending patterns — before money moves, not weeks later in a random audit sample.
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Key Deliverables	● Duplicate and split-transaction detection across time windows and submitters ● Behavioral anomaly model comparing spend against peer group, role, location, and historical patterns ● Document forensics to detect AI-generated, edited, or tampered receipt images and PDFs ● Per-report risk scoring feeding a prioritized audit queue in place of random sampling ● Case management workflow for the finance/audit team to investigate flagged items
Technical Approach	Statistical/ML anomaly detection (e.g., isolation forest) on structured transaction data, combined with Claude-based document forensics and contextual reasoning, plus hard-signal rules (duplicate hashes, sequential invoice numbers).
Timeline	Weeks 11–14
Success Metrics	● Audit sampling shifts from random to 80%+ risk-weighted ● Measurable improvement in fraud/error catch rate vs. baseline ● False-positive investigation rate under 10%
Key Risks	● Anomaly models need historical data to calibrate well ● Risk of biased flagging across employee groups — requires a fairness review before go-live

Phase 4 — Workflow Automation & Approvals

Objective	Automate the end-to-end approval and reimbursement workflow so compliant expenses move with minimal human touch, and exceptions route to the right person quickly.
Key Deliverables	● Dynamic approval routing (manager, cost-center owner, finance) based on amount, category, and risk score ● Zero-touch auto-approval path for policy-compliant, low-risk expenses ● Escalation and SLA tracking for pending approvals, with automated reminders ● Reimbursement trigger to payroll or AP payment rails once approved ● Full audit trail — who, what, when, why — for every automated decision
Technical Approach	A workflow/state-machine orchestration layer consumes the rules-engine output and risk scores from Phases 2–3; a Claude agent handles exception routing and drafts concise approval summaries for human approvers.
Timeline	Weeks 15–18
Success Metrics	● Reimbursement cycle time reduced from baseline (target: days, not weeks) ● 70%+ zero-touch approval rate ● SLA adherence above 95%
Key Risks	● Timing dependencies with payroll cycles ● Approver adoption and change management

Phase 5 — Enterprise System Integration

Objective	Connect the platform to systems of record so data flows automatically in both directions, avoiding the one-way-feed drift that plagues legacy expense tools.
Key Deliverables	● Bidirectional ERP/GL integration (SAP, Oracle, NetSuite, Workday, Dynamics) for GL coding, cost-center posting, and reconciliation ● HRIS integration for employee, role, manager, and cost-center data ● Card program and bank feed integration for real-time transaction matching ● Payroll/AP integration for reimbursement disbursement ● API layer with webhook support for real-time events, plus a batch fallback for legacy systems
Technical Approach	Native connectors where available, middleware/iPaaS to bridge gaps, idempotent sync design, and a reconciliation job that catches and alerts on drift between the platform and the GL.
Timeline	Weeks 19–22
Success Metrics	● Reconciliation discrepancy rate under 1% between expense platform and GL ● Integration uptime above 99.5%
Key Risks	● Client-specific ERP customizations ● Legacy systems with limited or no API support

Phase 6 — Employee Experience & Analytics

Objective	Make the system genuinely easy to use and give finance leadership real visibility — closing the loop that best-in-class tools emphasize: self-service, transparency, and spend insight.
Key Deliverables	● Employee self-service portal/mobile app to submit, track status, and see policy explanations in plain language ● Claude-powered conversational assistant for "why was this flagged," "what's my per-diem for this trip," and general policy Q&A ● Finance and manager dashboards: spend by category/department/trend, policy exception trends, cycle-time metrics ● Proactive nudges — e.g., submit-before-trip-ends reminders, missing-receipt alerts
Technical Approach	A Claude-powered assistant grounded strictly in the policy rulebook and the employee's own transaction data (to prevent hallucinated policy answers), paired with an analytics layer over the structured transaction store.
Timeline	Weeks 23–25
Success Metrics	● Employee satisfaction (CSAT) improves vs. baseline ● Self-service resolution rate for status/policy questions above 80%
Key Risks	● Assistant giving incorrect policy answers if not tightly grounded — requires strict guardrails and citation of the source rule

Phase 7 — Pilot, Hardening & Rollout

Objective	Validate the system with a real population before full rollout, and harden it for scale, security, and compliance.
Key Deliverables	● Pilot with one business unit, run in parallel with the existing process to compare accuracy and cycle time ● Security review, penetration testing, and SOC 2/GDPR compliance sign-off ● Performance and load testing for peak periods (month-end, quarter-end) ● Training materials and a change-management rollout plan ● Phased company-wide rollout plan with defined rollback criteria
Technical Approach	A/B comparison against the legacy process, an incident-response runbook, and monitoring/alerting on model drift and extraction accuracy over time.
Timeline	Weeks 26–30
Success Metrics	● Pilot shows measurable improvement in cycle time, accuracy, and audit coverage vs. baseline ● Zero critical security findings open at go-live
Key Risks	● Pilot scope creep ● Insufficient training-data diversity across departments and geographies

Phase 8 — Continuous Improvement (Post-Launch)

Objective	Keep the system accurate and aligned with evolving policy, fraud patterns, and business needs after go-live.
Key Deliverables	● Quarterly policy rule reviews ● Model recalibration using new fraud/anomaly data and audit-team override feedback ● Expansion to new geographies, currencies, and subsidiaries ● Feedback loop from audit outcomes back into the detection models
Technical Approach	Scheduled retraining/recalibration cadence with a feedback loop that captures every audit-team override to improve model precision over time.
Timeline	Ongoing, quarterly cadence
Success Metrics	● Sustained or improving accuracy and fraud-catch rates quarter over quarter
Key Risks	● Policy drift if reviews are skipped ● Model staleness without a feedback loop

Note on the First-Cut Prototype

A first-cut prototype (built separately, using Claude) will demonstrate a thin vertical slice of Phases 1–2: upload a receipt → Claude extracts structured data → the policy engine validates it against a sample rulebook → an

approval recommendation with a plain-language explanation is returned. This proves the core extraction-plus-reasoning loop end to end before the larger integration and workflow phases are built out.